



ULTRA AVIATION SERVICES, INC.

General Operations Manual

SECTION K

GROUND SERVICE EQUIPMENT (GSE)

MANAGEMENT



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Revision: 6
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Section: K GROUND SERVICE EQUIPMENT (GSE) MANAGEMENT

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Date: 10/01/2021**K.2 GSE MAINTENANCE POLICY****ORM 2.1.1, 2.1.2**

Any ground service equipment utilized by the company to perform ground handling services shall be maintained under a maintenance program that ensures that GSE remains safe to operate and in good condition, and:

- Is maintained in accordance and/or guidance from the GSE manufacturer or under established company fleet management policies.
- Is serviceable and in good condition prior to being used in ground operations, and such that it will not endanger vehicle users, other vehicles, pedestrians, aircraft or property.
- When found to be defective, is reported and evaluated for removal from service;
- Tagged as "Out of Service" when removed from operations;
- Maintenance records are retained for a minimum period of one (1) year, unless otherwise required by applicable regulations or customer airlines.

The company will provide staff, facilities and other resources as necessary to the person responsible for GSE maintenance.

K.3 GSE MAINTENANCE PROGRAM

The Company Maintenance program is composed of the following components.

- **Inspections** - Daily Walk-Arounds, Visual and Functional inspections are important in order to maintain awareness of the condition of ground service equipment being used to conduct ground handling service operations.
- **Scheduled Maintenance** or Preventive Maintenance is a key component of maintaining reliable ground service equipment. The company has developed its programmed maintenance interval (PMI) table with the GSE manufacturer's guidance, industry best practices, and inline with customer airline(s) requirements.
- **Un-Scheduled Maintenance** is necessary to return ground service equipment to service quickly in order to avoid any interruption to ground handling operations.
- **Audits** - Periodically, Audits will be performed both internally and by customer airlines in order to ensure that the GSE Maintenance Program is being followed.



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Date: 10/01/2021**K.3.1 PROGRAMMED MAINTENANCE INTERVALS**

All Company Ground Service Equipment (GSE) that is in service while conducting ground handling operations shall be maintained according to the following interval table.

Preventive Maintenance Interval (PMI) Table

EQUIPMENT TYPE	MOTORIZED	DAILY	MONTHLY	QUARTERLY	ANNUALLY
		WALK AROUND	SAFETY INSPECTION	"A" CHECK	"B" CHECK
TUG / BAGGAGE TRACTOR	X	X	X	X	X
PUSHBACK TRACTOR	X	X	X	X	X
BELT LOADER	X	X	X	X	X
CONTAINER LOADER	X	X	X	X	X
LAVATORY TRUCK	X	X	X	X	X
WATER TRUCK	X	X	X	X	X
AIR STAIRS	X	X	X	X	X
HIGH LIFT TRUCK	X	X	X	X	X
GROUND POWER (GPU)		X		120 hrs	1500 hrs
AIR CONDITIONER (ACU)		X		120 hrs	1500hrs
AIR START (ASU)		X		120 hrs	1500hrs
WATER CART		X	X		X
LAVATORY CART		X	X		X
CAR/VAN/TRUCK	X	X	X	X	X
GOLF CART		X		X	
TOW BAR		X	X	X	
CARTS / DOLLIES		X	X		
AIR STAIRS (Non-Motorized)		X	X		
MAINTENANCE RECORDS					AUDIT

- **DUE DATES:** Quarterly and Annual Maintenance is considered to be due at the end of the month in which the due date falls.
- **GRACE PERIOD:** A 30 DAY GRACE PERIOD will be granted on all quarterly and annual checks during which the piece of ground service equipment may remain in services, provided that the weekly safety inspections are current and no discrepancies have been found. If the required maintenance has not been performed by the end of this grace period, then the piece of ground service equipment must be removed from service until the required maintenance has been accomplished and the GSE has been deemed fit to return to service.

**K.3.2 GSE INSPECTIONS**

Prior to putting any piece of Ground Service Equipment (GSE) in service a safety and functionality inspection must be performed. Once in service GSE Safety Inspections should be performed as follows:

Daily	- At the beginning of their shift, each Ramp Supervisor must perform a safety inspection of the equipment that they will be using in ground handling operations under their supervision, and report any discrepancies to the Ramp Manager, and the GSE Department or GSE vendor. - On a daily basis, the GSE Department or GSE vendor must check with the Ramp Manager to address any discrepancies.
Monthly	On a monthly basis, the GSE department or GSE vendor must perform and document a GSE safety inspection of all ground service equipment that is in use in conducting ground handling operations.

A. GSE SAFETY INSPECTION ITEMS

As applicable to the type of ground service equipment, check the following items for before placing operating a piece of GSE:

Motorized Equipment

- Current Airport Decal
- Fire Extinguisher
- Brake Lights
- Headlights
- Tail Lights
- Tire Condition
- Tire Pressure
- Fluid Leakage
- Rubber Bumpers
- Physical Damage
- Windshield / Windows
- Tow Hitch / Connector
- Controls / Steering
- Brake Test
- Engine Functionality
- Transmission Test
- Test Drive
- Safety Systems
- Seat Belt

Additional Items for GSE Department / Vendor

- Fluid Levels
- Tire Nuts
- Battery
- Company Decals

Non-Motorized Equipment

- Tire Condition
- Tire Pressure
- Tow Hitch / Connector
- Stabilizers
- Locks / Curtains
- Equipment Functionality
- Brakes

Tow Bars

- Body
- Hydraulic System
- Shear Bolts
- Hub Wheel and Tire
- Lock Pin
- Lunnette Weld



B. UNSERVICEABLE GSE

Ground Service Equipment with defective brakes, protective buffers (bumpers) steering tow couplings, lights, wheels or windshield wipers or with fuel or oil leaks must be tagged and removed from service and not utilized in airside operations.

Adopt the following procedures when unserviceable equipment needs to be removed from the operations area:

1. Tag the equipment as "Out of Service".
2. Notify the supervisor, who should contact the maintenance provider.
3. Remove the vehicle from the area according to the emergency breakdown procedures.
4. Do not operate vehicles or equipment labeled unserviceable until they have been repaired, inspected and the supervisor has released the equipment back into service.



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Scheduled Maintenance is the key to maintaining reliable ground service equipment. The following items are to be accomplished and documented at the intervals indicated in the PMI table in this section:

A. "A" CHECK - QUARTERLY**• PRE-OPERATION AND DRIVE TEST**

1. Review History
2. Do a visual inspection for leaks and wash unit
3. Perform walk around inspection (chock wheels)
4. Note General Condition of Company Decals on the unit
5. Check Fire Extinguisher, Safety Equipment and decals.
6. Check engine fluid levels (Fuel, Engine Oil and Coolant).
7. Check Parking Brake and Service Brake
8. Check lights and lens, dash, running and turn beacons.
9. Check Seat, Seat Belt and Seat Adjustment tracks.
10. Check Choke Cable and Horn.
11. Start Unit: Check Gauges (Oil, Voltage, Temperature, Engine Noise)
12. Check for excessive smoke, misfires, gauge response.
13. Check steering wheel, steering column, and steering play.
14. Test Drive and Transmission operation
15. Test Brake performance (service brake pull)
16. Pedals, pads and shift lever
17. Check transmission fluid.

• ENGINE COMPARTMENT

1. Check Ignition, charging and starting system.
2. Check Master cylinder lever and booster
3. Inspect Belts and Hoses for condition and adjustment
4. Check Water Pump, Fan and Alternator
5. Check Battery: Cables, Connections and Acid levels
6. Check Fuel Pump and Hoses.
7. Check Fuses, Breakers, and Wiring.
8. Check Air Filter and Replace as Required.
9. Check Intake System for leaks.
10. Check heat riser and choke heat tube.
11. Change Oil and Oil Filter
12. Replace Fuel Filter.



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- **UNDERCARRIAGE AND BODY**

1. Check Front Axle Assembly and Suspension.
2. Check Front Wheel Bearings and King Pins
3. Inspect Front Brakes for wear and leaks and Replace as required
4. Check Steering Gear mounting and oil level.
5. Check Steering Linkage, Cotter Pins and Lock Bolts.
6. Inspect Frame, Cross Members and Engine Mounts.
7. Check condition of the exhaust system and hangers.
8. Inspect Body Bolts, Electrical Cables and Brake Lines.
9. Check Transmission: (linkage, mounts and leaks)
10. Inspect Drive line, U-joints and yokes
11. Check Rear Suspension Mounts
12. Check Differential Fluid and Level (Clean Breather)
13. Check Parking Brake Assembly and Adjustment
14. Check Rear Brakes and Lines and Replace as required.
15. Check Tire Pressure (adjust as required)
16. Check Condition of tires, rims, lug nuts and wheels
17. Inspect Rear hitch and safety lock

- **LUBRICATION**

1. Lubricate Thoroughly

- **APPEARANCE**

1. Report any damage or paint defects

- **CUSTOMER AIRLINE(S)**

1. Additional Checks and Inspections as required by Customer Airlines(s)

B. "B" CHECK - ANNUALY

- Perform all items included in "A" Check in addition to the following items:
 1. Clean and repack wheel bearing.
 2. Clean and replace PVC valves as required
 3. Service transmission and filters
 4. Remove Battery
 5. Clean all battery terminals.
 6. Clean battery box and paint as needed.
 7. Do major tune up
 8. Check governor
 9. Change differential oil and breather.
 10. Flush radiator and replace coolant.



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Un-Scheduled Maintenance is required when a GSE unit is reported to be unserviceable. Unserviceable equipment can cause interruptions to ground service operations, damage to aircraft, and create unsafe conditions for employees. For these reasons, when a piece of ground service equipment becomes inoperable it is extremely important that the unit is reported to the Ramp Manager and the GSE Maintenance Department or Vendor is notified so that it can be returned to service as soon as possible.

If a piece of Ground Service Equipment (GSE) is not functioning properly, adopt the following procedure.

1. Check the Fuel Quantity.
2. Report the unit to the Ramp Supervisor and/or Manager.
3. Notify the GSE Maintenance Department and/or Vendor.
4. Conduct a Safety Inspection
5. If the unit can be fixed with minimal effort, then correct the issue and return the unit to service.
6. If the unit is unserviceable, then tag the unit as "Out of Service" and remove it from the area of ground handling operations until corrective actions have been made, the unit has been inspected and can be returned to service .
7. Unless operational conditions do not allow, If the unserviceable unit is due for scheduled maintenance, then perform the programmed maintenance prior to returning the unit to service

K.3.5 AUDITS

An internal audit of maintenance records and inventory shall be conducted by the company quality assurance department on at least once every 13 months in order to ensure that the program is being followed as required. Additional audits may be conducted by customer airlines periodically as required.



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K.4 GROUND SERVICE EQUIPMENT (GSE) TECHNICAL REQUIREMENTS

ORM 2.2.1

All aircraft ground support equipment (GSE) **should**:

- i. comply with organizational, national, local and airport requirements for its intended use.
- ii. is only used for the purpose of its designed purpose, including for specific aircraft types.
- iii. have appropriate standard operational procedures and,
- iv. human factor principles regarding the controls of the vehicle are considered during procurement.

K.4.1 PROXIMITY SENSING AND WARNING DEVICES

ORM 2.2.2

It is recommended that:

- i. any new self-propelled GSEs should be equipped with proximity sensing and warning devices that provide the capacity to sense the proximity of an aircraft and provide a visual and audible indication to the operator of the GSE's position, to reduce the risk of impact with the aircraft.
- ii. the environmental cost of new equipment is considered during procurement, and that low or zero emissions technology is chosen where feasible.
- iii. if GSE is acquired by the company which is equipped with proximity sensing and warning devices, then the company should perform and update a risk assessment, update any related operational procedures and provide the necessary training to relevant personnel.

If economically possible, current self-propelled GSEs should be retro-fitted with proximity sensing and warning devices.



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Date: 02/02/2022**K.5 POTABLE WATER****HDL 1.5.4**

Access to safe drinking water is essential to health, a basic human right and a component of effective policy for health protection. One of the significant risks to aviation is posed by the potential for microbial contamination of aircraft water by animal or human excreta. This contamination may originate from source waters or may occur during transfer operations or while water is stored on board the aircraft. However, any location is at risk if proper procedures and sanitation practices are not continuously followed to ensure the safety of water that is used for drinking and food processing and preparation. It is therefore essential that the potable water is disinfected and that handling companies adhere to sanitary requirements.

Even if the water at the airport is safe, that does not ensure that it will remain safe during the transfer to the aircraft and storage activities that follow. An understanding of the aircraft drinking water supply and transfer chain will help to illustrate the points at which the water can become contaminated en route to the tap on board the aircraft. Generally, the aircraft drinking water supply and transfer chain consists of four major components:

- (a) The source of water coming into the airport
- (b) The airport water system, which includes the on-site distribution system. It may also include treatment facilities if the airport produces its own potable system.
- (c) The water transfer point and delivery system between the hard-plumbed potable water cabinet provided by the airport or the potable water service unit used by the ground service provider, and the aircraft water system. This water transfer process provides multiple opportunities for the introduction of contaminants into the drinking water.
- (d) The aircraft water system, which includes the water service panel, the filler neck of the aircraft finished water storage tank and all finished water storage tanks, including refillable containers/urns, piping, treatment equipment and plumbing fixtures within the aircraft that supply water to passengers or crew.

As a ground service provider the company is responsible ensuring that no contaminated water is uploaded into aircraft being serviced at the water transfer point between the potable water service unit and the aircraft and that water quality standards, preparation, handling and inspection procedures are in place in accordance with local health authorities and the customer airline.



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Date: 02/02/2022**K.5.1 WATER QUALITY STANDARDS****HDL 1.5.4**Water Testing Schedule

- **Water Truck** - Each water service unit will be tested on a bi-weekly basis by the company GSE Maintenance Department and a record of each test retained.
- **Airport Cabinet Test** - The company shall test the water source from which the water service unit is filled on a monthly basis by the company GSE Maintenance Department and a record of each test retained.
- **Laboratory Test** - Each water service unit will be tested on a quarterly basis and sent to a qualified laboratory for analysis. A record of the result shall be retained.

COMPANY CONDUCTED TESTWater Sample Test Preparation

- Use the recommended personal protective equipment. (Rubber Gloves)
- Analyze samples immediately after collection.
- Put the color disc on the center pin in the color comparator box (numbers to the front).
- Use sunlight or a lamp as a light source to find the color match with the color comparator box.
- Rinse the tubes with sample before the test. Rinse the tubes with deionized water after the test.
- If the color match is between two segments, use the value that is in the middle of the two segments.
- If the color disc becomes wet internally, pull apart the flat plastic sides to open the color disc.
- Remove the thin inner disc. Dry all parts with a soft cloth. Assemble when fully dry.

Undissolved reagent does not have an effect on test accuracy.

The glycine reagent removes 6 mg/L chlorine from the sample.

Water Sample Test Procedure

1. Fill two tubes to the first line (5ml).
2. Put one tube into the left opening of the color comparator box.
3. Add 2 drops of glycine reagent to the second tube.
4. Swirl to mix.
5. Add one DPD Free Chlorine Reagent Powder Pillo to the same tube.
6. Swirl to mix. A pink color develops.
7. Put the second tube into the color comparator box.
8. Hold the color comparator box in front of a light source. Turn the color disk to find the color match.
9. Read the result in mg/L in the scale window.
10. The Free Chlorine test result shall be 0.3 to 0.8 mg/l. If peroxide is added for disinfection, the result shall be 0.1 to 0.3 mg/l H₂O₂. If the result is not within limits, re-sample and test again.
11. If a second failed test is recorded, then drain the tank completely, refill and test again.
12. Record the results on the company results form and in the water truck maintenance log.



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Schedule a sample collection appointment by phone or email with the laboratory company and they will collect a water sample and return the results.

Company: Applied Consumer Services, Inc.**Email:** office@appliedconsumer.com**Phone:** 305-821-1677

Retain a copy of the results in the water truck maintenance file.



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Date: 02/02/2022**K.5.2 DRINKING WATER VEHICLE MAINTENANCE****A. VEHICLE CONDITION**

- Vehicles shall be maintained to a generally accepted standard of mechanical reliability, safety and the water tanks shall be leak free. This will cover all daily and weekly serviceability checks through to scheduled preventative maintenance of engine, chassis and pumping/servicing equipment.
- Ensure that tank roof area water drains (if any) are clear.
- Ensure that tank lid/dome cover gaskets and tank vents are in good condition and operate correctly.
- Vehicles shall be in a roadworthy condition.
- Vehicles shall be clearly identified as "Drinking/Potable Water for Aircraft" vehicles.
- Filters change due date shall be clearly indicated.
- Hoses shall be of food and beverage quality. Hoses shall not have cracks and/or kinks. Maximum lifespan shall not exceed three years or as per manufacturers specifications.
- Hoses connectors shall have flat connecting valves and not be worn. Intermediate connector adaptors are prohibited.
- Hoses shall be capped or secured with a dummy connector. Hose end dust caps shall be clean, in good condition and in place when hoses are not in use.
- The hose connectors must be water-proof or at least food grade quality.
- Water tanks shall be designed so that they can be disinfected and flushed and shall be provided with a drain that permits complete drainage of the tank. The drainage port shall not leak, and shall be located at the lowest point of the tank to allow for full drainage. The drainage port shall be closed after the tank has been drained.

B. DISINFECTION OF VEHICLES

- Servicing of vehicle potable water tanks and hoses shall include disinfection on a bi-weekly basis in accordance with the disinfection/cleaning procedures. The last/next cleaning disinfection dates shall be clearly displayed.
- Vehicle cleaning - Water tank interiors shall be cleaned internally at least once a month to remove any scale or deposit. Rust and lime scale inside of water bowsers can contaminate the water.
- Visually inspect the tank(s) through the upper opening (if applicable) to ensure that the internal surfaces are well maintained and clean.



Tank Disinfection Procedure

1. Perform Chlorine test, if results are not adequate, drain the water tank completely
2. Fill the tank with 1000 gallons of water and 0.25 gallons of chlorine to the tank.
3. Drive the truck around for 10 minutes to stir the liquid and sanitize the tank.
4. Drain all of the liquid from the tank.
5. Fill and Drain the tank 3 times to rinse the tank and remove all excess chlorine.
6. Fill the tank with water provided by the county.
7. Perform Chlorine Test, if results are not adequate, drain the water tank completely and repeat.
8. Once the required results have been obtained, document that the procedure was performed and retain a record in company files.

K.5.3 DRINKING WATER VEHICLE RECORDS

- Maintenance records will be kept up-to-date to ensure vehicles are maintained in good condition for safe and reliable service. The records shall contain the following as a minimum:
 - Daily checks (refilled within 24 hours and chlorine or other disinfectant content);
 - bi-weekly disinfection records;
 - monthly cleaning records;
 - Laboratory Analysis;
 - Details of work carried out, including servicing, repairs and replacement parts;